# Liver Disease — Hepatitis, Cirrhosis and NAFLD

## 1. Overview

The liver is the body's largest internal organ, performing over 500 metabolic functions including protein synthesis, gluconeogenesis, lipid metabolism, bile production, detoxification, and immune function. Liver disease encompasses a spectrum from mild elevation of liver enzymes to liver failure and cancer.

In India, liver disease is a significant public health problem. Viral hepatitis (B and C) accounts for the largest share of liver disease burden globally. Non-alcoholic fatty liver disease (NAFLD), now reclassified as Metabolic Associated Fatty Liver Disease (MAFLD), is the fastest growing liver disease, driven by obesity, diabetes, and the metabolic syndrome.

## 2. Viral Hepatitis

Hepatitis A and E: faeco-oral transmission (contaminated water, food). Self-limiting acute hepatitis. Hepatitis A: worldwide, children in endemic areas. Hepatitis E: major cause of acute hepatitis in India, particularly during floods/monsoon. Very high mortality in pregnancy (15-25% — fulminant liver failure). No specific treatment.

Hepatitis B (HBV): blood-borne (IV drug use, needlestick, unprotected sex) and vertical (mother to child at delivery). 300 million chronically infected worldwide, 40+ million in India (3.7% prevalence). Leads to chronic hepatitis, cirrhosis, and hepatocellular carcinoma (HCC). HBsAg is the surface antigen used for diagnosis; HBeAg indicates high infectivity; HBV DNA quantifies viral load.

Treatment of chronic HBV: tenofovir disoproxil fumarate (TDF) or entecavir (ETV) — preferred first-line oral antivirals. Suppress viral replication, reduce risk of cirrhosis and HCC. Not curative — long-term or lifelong therapy often required. Pegylated interferon-alpha (Peg-IFN) is an alternative finite duration treatment.

Hepatitis C (HCV): blood-borne. 58 million chronically infected globally. India: 6-12 million estimated. Major cause of cirrhosis and HCC. HCV is CURABLE with Direct-Acting Antivirals

(DAAs): sofosbuvir-velpatasvir, sofosbuvir-ledipasvir, glecaprevir-pibrentasvir. 8-12 week treatment, cure rates >95%. Cost has fallen dramatically — affordable generics available in India.

## 3. Non-Alcoholic Fatty Liver Disease (NAFLD/MAFLD)

NAFLD is defined as >5% hepatic steatosis (fat in liver) without significant alcohol use. It is the most common liver disease globally, affecting approximately 25% of the world population. In India, NAFLD affects ~25-35% of the urban population, closely linked to the epidemic of metabolic syndrome.

Spectrum: Simple steatosis (NAFL) → Non-Alcoholic Steatohepatitis (NASH — steatosis + inflammation + injury) → Fibrosis → Cirrhosis → HCC. Approximately 20% of NAFLD progresses to NASH; 20-30% of NASH progresses to cirrhosis over 10 years.

Diagnosis: Liver ultrasound (first-line screening). FibroScan (transient elastography) measures liver stiffness (fibrosis). NAFLD Fibrosis Score (NFS) or FIB-4 index (non-invasive scoring using routine bloods). Liver biopsy — gold standard for staging fibrosis and confirming NASH.

Treatment: no approved medications for NAFLD/NASH yet (as of 2024). Cornerstone: weight loss (7-10% body weight reduces steatosis, >10% improves fibrosis). Diet: Mediterranean diet. Exercise: both aerobic and resistance. Control metabolic risk factors. SGLT2 inhibitors and GLP-1 agonists (semaglutide, liraglutide) show promising results in trials. Resmetirom (THR-beta agonist) FDA approved 2024 for non-cirrhotic NASH with fibrosis.

## 4. Alcoholic Liver Disease

Alcoholic Liver Disease (ALD) encompasses fatty liver (reversible), alcoholic hepatitis (potentially severe and life-threatening), and alcoholic cirrhosis. Heavy drinking (>4 standard drinks/day for men, >2 for women) over years is required.

Alcoholic hepatitis: acute presentation with jaundice, fever, hepatomegaly, tender liver, leukocytosis. Discriminant Function (Maddrey) ≥32 or MELD >20 = severe — prednisolone 40 mg/day x 28 days reduces short-term mortality. Pentoxifylline is an alternative. Liver transplant for selected cases not responding to steroids (excellent outcomes).

Abstinence is the most important intervention at all stages of ALD. Even established cirrhosis can compensate and 5-year survival improves dramatically with complete abstinence.

## 5. Cirrhosis

Cirrhosis is the end-stage of chronic liver disease — extensive fibrosis distorting normal liver architecture with formation of regenerative nodules. Leads to impaired liver function and portal hypertension.

Compensated vs Decompensated: compensated cirrhosis — no major complications, may be asymptomatic. Decompensated — ascites, spontaneous bacterial peritonitis (SBP), variceal bleeding, hepatic encephalopathy, hepatorenal syndrome (HRS).

Ascites: most common complication. Na restriction (<2g/day) + spironolactone (100-400 mg) + furosemide (40-160 mg). Therapeutic paracentesis for large-volume ascites. TIPS (transjugular intrahepatic portosystemic shunt) for refractory ascites. Avoid NSAIDs and nephrotoxins.

Hepatic encephalopathy: altered consciousness due to uraemic toxins (ammonia) reaching brain. Lactulose (titrated to 2-3 soft stools/day) is first-line. Rifaximin (non-absorbable antibiotic) prevents recurrence. Identify and treat precipitants: GI bleed, infection, constipation, dehydration, sedatives.

Variceal bleeding: medical emergency — IV terlipressin/octreotide, broad-spectrum antibiotics (prophylactic), endoscopic variceal band ligation (EVL) within 12 hours. Non-selective beta-blockers (propranolol, carvedilol) for primary prevention of variceal bleeding.

## 6. Hepatocellular Carcinoma (HCC)

HCC is the most common primary liver cancer and the third leading cause of cancer death globally. Approximately 80% arise in cirrhotic livers. Major risk factors: HBV, HCV, alcoholic cirrhosis, NASH cirrhosis, aflatoxin B1 exposure (food contamination — maize, groundnuts in India).

Surveillance: all cirrhotic patients and HBV carriers (without cirrhosis but with risk factors) should undergo 6-monthly liver ultrasound ± AFP for HCC surveillance.

Staging: Barcelona Clinic Liver Cancer (BCLC) system. Treatment: curative options (resection, liver transplant, radiofrequency ablation/RFA) for early-stage. TACE (transarterial chemoembolization) for intermediate. Sorafenib, lenvatinib (targeted therapy), atezolizumab-bevacizumab (immunotherapy combination) for advanced.

## 7. Acute Liver Failure

Acute liver failure (ALF): rapid deterioration of liver function in a patient without pre-existing liver disease — coagulopathy (INR >1.5) + encephalopathy within 26 weeks of illness onset.

Causes in India: HAV (most common in children), HEV (most common in adults, especially pregnancy), HBV, drugs (antitubercular drugs, paracetamol overdose, herbal remedies with hepatotoxins).

Management: ICU care. Lactulose + rifaximin for encephalopathy. Vitamin K, fresh frozen plasma for coagulopathy (if bleeding). Paracetamol-induced ALF: N-acetylcysteine IV (highly effective if given within 8-16 hours). Monitor for cerebral oedema, renal failure, hypoglycaemia, infections.

Emergency liver transplant: the only definitive treatment for ALF meeting King's College Criteria. Transplant within 72 hours dramatically improves survival. Extremely limited availability in India.

## 8. Hepatitis B Vaccination and Prevention

Hepatitis B vaccine is one of the most important vaccines. The universal childhood immunisation schedule in India includes HBV vaccine at birth (within 24 hours for MTCT prevention), 6 weeks, 10 weeks, and 14 weeks.

Catch-up vaccination: all unvaccinated adolescents and adults at risk (healthcare workers, sexual contacts of HBsAg-positive individuals, IDUs, dialysis patients). Three doses (0, 1, 6 months) or accelerated schedule.

Prevention of hepatitis C: no vaccine available. Harm reduction for IDUs (needle exchange, OST). Screen blood donors. Safe sex practices. Universal precautions in healthcare settings. Screen all PLHIV and high-risk individuals for HCV.

Aflatoxin prevention: proper grain storage (dry, ventilated), regulatory testing of food commodities, avoiding consumption of visibly mouldy food — important HCC prevention measure in India.

## 9. Drug-Induced Liver Injury (DILI)

DILI is the most common cause of acute liver failure in developed countries and a significant problem in India. Antitubercular drugs (isoniazid, rifampicin, pyrazinamide) are the most common cause of DILI in India — occurring in 5-28% of patients on standard RIPE therapy.

Other important causes in India: herbal and traditional medicines (Ayurvedic preparations containing heavy metals, herbs with pyrrolizidine alkaloids), antiretrovirals (nevirapine, efavirenz), complementary and alternative medicines, dietary supplements.

Monitoring: baseline LFTs before starting ATT. Repeat at 2 and 4 weeks, then monthly. Stop ATT if ALT >5x ULN or ALT >3x ULN with symptoms/jaundice. Introduce drugs sequentially when reintroducing after DILI resolution.

## 10. Liver Health in India — Prevention

Hepatitis B: universal infant vaccination with zero dose at birth within 24 hours is the most effective intervention against HBV-related cirrhosis and HCC. Safe injection practices and blood safety.

Hepatitis C: India is scaling up DAA-based treatment through the National Viral Hepatitis Control Programme (NVHCP) — targets treating all HCV-infected individuals. Free Sofosbuvir + Daclatasvir provided at government hospitals.

NAFLD prevention: population-level interventions against metabolic syndrome — reducing consumption of sugar-sweetened beverages, promoting physical activity, reformulation of processed food, school-based nutrition programmes.

Alcohol: strengthening alcohol policy — taxation, restricting advertising, improving access to treatment for AUD, support for community-based de-addiction programmes.

Food safety: enforcement of mycotoxin (aflatoxin) limits in food commodities, particularly groundnuts and maize. Awareness about safe food storage practices in homes and small businesses.